Diagnosing Swapped Input–Output and Function-Notation Errors

Directions. Function notation packs two different numbers into one short expression: in $f(3) = 7$, the 3 is the **input** and the 7 is the **output**. Almost every mistake below comes from putting a number in the wrong one of those two slots. For each problem, decide first *which slot the given number belongs in*, then do the arithmetic. Where a student's work is shown, name the mistake in words before you fix it.

1. Let $f(x) = 4x - 5$. Find $f(3)$. Before you compute: is 3 an input or an output?

Answer: ________________

2. Still with $f(x) = 4x - 5$. Find the input x for which $f(x) = 19$. This time the 19 sits on the output side, so substituting it for x would answer a different question.

Answer: ________________

3. The table gives $g(m)$, the number of rainy days recorded in month m at one weather station.

m	1	2	3	4	5
$g(m)$	4	1	5	2	3

Priya says $g(2) = 4$. She found the 2 in the bottom row and read the month above it. Explain in one sentence what Priya's method actually computes, then write the correct value of $g(2)$.

Answer: ________________

4. Jo is working with $f(x) = 4x - 5$ and writes

$$f(2 + 3) = f(2) + f(3).$$

Jo is reading $f(\ )$ as though it means "f times whatever is inside." Compute $f(2 + 3)$ correctly, and state the value Jo's method would give instead.

Answer: ________________

5. A taxi company charges a flat \$3.00 plus \$2.50 for each mile driven, so the cost in dollars for m miles is $C(m) = 3 + 2.5m$. Kim has \$28.00 and wants to know how many miles she can ride. She starts by writing $C(28)$. Explain why $C(28)$ answers the wrong question, then find the number of miles Kim can ride.

Answer: ________________

6. Use the same station data as problem 3, where $g(4) = 2$, together with a second function $B(x) = 2x - 3$. Theo claims "g is the bigger function because its table contains a 5." Compare the two functions properly at the input 4: write $g(4)$ and $B(4)$, then state which is greater using $<$ or $>$.

Answer: ________________

7. Marcus was asked: "For $h(x) = 30 - 6x$, find the input that gives an output of 12." Here is his work.

$h(12) = 30 - 6(12) = 30 - 72 = -42$
so the input is -42

Circle the first line where Marcus goes wrong, say what he swapped, and find the correct input.

Answer: ________________

8. Write a rule, in your own words, that a classmate could use to decide whether a question is asking them to **evaluate** a function (the input is given) or to **solve** for an input (the output is given). Then show your rule working on these two questions about $f(x) = 4x - 5$: "Find $f(3)$" and "Find x when $f(x) = 3$." Your two answers should be different numbers — explain why that is exactly the point.